

DHSNet: Denoised-Modulated Hybrid-Semantic Scale-Aware Network for Low-Light Image Enhancement

Rentao Yang, Zhize Wu, Xiaofeng Wang, Tong Xu, Fengling Jiang, Amir Hussain, and Le Zou

Abstract—Low-Light Image Enhancement (LLIE) methods based on either Retinex theory or deep learning still exhibit significant shortcomings in handling image corruptions, such as noise, artifacts, and color distortion. The primary issue is that both Retinex algorithms and existing networks may introduce or amplify these corruptions during enhancement. To address these limitations, we propose the Denoised-Modulated Hybrid-Semantic Scale-Aware Network (DHSNet), a novel one-stage LLIE method. DHSNet integrates a Signal-to-Noise Ratio (SNR)-based denoising mechanism and a Hybrid-Semantic Scale-Aware Module (HSM) to preprocess noise and fuse multi-scale features for robust image enhancement. Moreover, we introduce the Illumination Partial Attention Block (IPAB) to further improve illumination correction and nonlinear transformation capabilities. DHSNet effectively mitigates noise, preserves intricate details, and restores degraded structures. Extensive experiments on multiple LLIE datasets demonstrate that it outperforms state-of-the-art (SOTA) methods in both qualitative and quantitative metrics. Furthermore, DHSNet exhibits strong generalization in no-reference LLIE and low-light object detection tasks, underscoring its practical value for real-world applications.

Index Terms—Low-Light Image Enhancement, Transformer, Supervised Learning.

I. INTRODUCTION

Images captured in low-light conditions often exhibit low contrast, color distortion, and noise. These degradations not only impair visual perception but also hinder downstream tasks such as object detection, classification, and segmentation. The primary goal of Low-Light Image Enhancement (LLIE) is to enhance brightness and contrast, restore lost details, and ultimately improve visibility.

Traditional techniques, such as histogram equalization [1] and gamma correction [2], can boost overall brightness and contrast; however, they typically ignore local scene variations and the complexities of non-uniform illumination. Retinex-based methods [3] decompose images into reflectance and

illumination components, yet they frequently neglect noise and artifact removal, limiting their practical effectiveness.

Recent advancements in deep learning have given rise to LLIE methods built on convolutional neural networks (CNNs) and Transformer architectures. While CNN-based approaches excel at extracting local features such as textures and fine details, they often struggle to model long-range dependencies across the image. In contrast, Transformer-based methods such as Restormer [4] use channel-wise self-attention to capture global context more efficiently than full spatial attention. Hybrid methods such as Retinexformer [5] and Retinexmamba [6] fuse Retinex-based decomposition with Transformer or Mamba [7] modules, leading to improved enhancement performance. However, Retinex-based decomposition is inherently sensitive to noise, which can be amplified during enhancement. Additionally, existing encoder-decoder architectures still struggle to handle noise and artifacts at multiple scales. As shown in Figure 1, these limitations allow degraded details to pass through skip connections and reduce final enhancement quality. Addressing these challenges, namely effective noise removal, minimizing enhancement interference, and ensuring robust restoration within the network, remains a critical objective.

In this paper, we propose a novel method for LLIE called Denoised-Modulated Hybrid-Semantic Scale-Aware Network (DHSNet), which introduces a denoising mechanism based on Signal-to-Noise Ratio (SNR) and a Hybrid-Semantic Scale-Aware Module (HSM). Specifically, the SNR-based denoising mechanism preprocesses inherent noise in the input image, generating a denoised version that is then fused with the original input. This fusion improves network robustness and provides a preliminary enhancement while preserving fine details. The HSM employs a pyramid structure with recursive skip connections to extract and fuse multi-scale features across different semantic levels. It serves as a gating mechanism to guide illumination correction and effectively repair image degradation.

To further refine illumination correction, we propose an Illumination Partial Attention Block (IPAB), which enhances the model's nonlinear transformation capacity and improves illumination adjustments. By integrating HSM and IPAB within an encoder-decoder framework, DHSNet efficiently suppresses noise, extracts multi-scale features, and enhances image quality in a single stage. Due to its robust preprocessing and multi-scale awareness, DHSNet preserves fine details, mitigates noise, and restores degraded structures. Compre-

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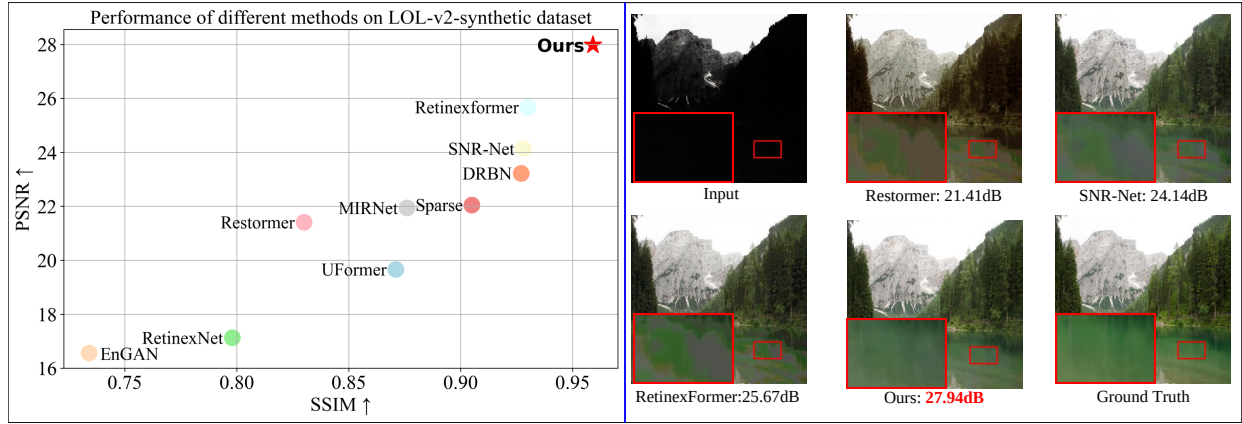


Fig. 1. Comparison of our method with other SOTA methods on the LOLv2-Synthetic dataset. The left panel presents quantitative results, while the right panel provides qualitative examples.

hensive qualitative and quantitative experiments demonstrate that DHSNet consistently outperforms SOTA methods across multiple LLIE datasets. Moreover, DHSNet shows exceptional performance in no-reference enhancement tasks and low-light object detection, significantly improving results on high-level vision tasks.

Our contributions are summarized as follows:

- We introduce DHSNet, a novel one-stage LLIE framework that suppresses noise and processes multi-scale features across different semantic levels to enhance image quality and restore degraded structures.
- We design a SNR-based denoising mechanism that effectively removes local noise, improves network robustness, and enables preliminary image enhancement via a modulation strategy.
- We develop HSM and IPAB: HSM fuses multi-scale, multi-semantic features to guide illumination correction and preserve fine details, while IPAB enhances nonlinear transformation capabilities and refines illumination adjustments.
- We validate DHSNet through extensive experiments on multiple LLIE datasets and low-light object detection tasks, demonstrating its performance and generalization capability compared to existing SOTA methods.

The remainder of this paper is structured as follows: **section II** reviews related works, **section III** presents our method, and **section IV** details the experimental results. Finally, we summarize our findings, discuss the limitations of our model, and outline potential directions for future work in **section V**.

II. RELATED WORK

A. Low-Light Image Enhancement

The primary objective of Low-Light Image Enhancement (LLIE) methods is to improve the visual quality of images captured under poor illumination. Traditional approaches, such as histogram equalization [8] and gamma correction [9], rely on global adjustments and often fail to adaptively address local features, noise, and varying illumination. As a result, they tend to produce noticeable noise, artifacts, and color distortions.

Moreover, these methods are highly sensitive to parameter settings and lack flexibility across diverse scenes. Retinex-based methods [10], [11] attempt to enhance images by decomposing them into reflectance and illumination components. However, they frequently overlook the impact of noise and color distortion, which can lead to further image degradation and amplification of artifacts. To overcome these limitations, this paper aims to improve the network's capacity to perceive and address such issues effectively.

With the rapid advancement of deep learning, numerous novel approaches have been proposed for LLIE. Wavelet-based methods [12], [13], [14] effectively model spatial domain features while enhancing image quality. Ye et al. [15] introduce an external memory-augmented network (EMNet), which leverages stored knowledge from the training data to guide the enhancement process during inference. Ma et al. [16] present a Retinex-based variational model that produces noise-free enhancement results. Yu et al. [17] propose an efficient baseline model, ESDNet, to address scale variation caused by moiré patterns in 4K images. Wang et al. [18] propose an adaptive enhancement framework for a single low-light image based on the strategy of virtual exposure, which effectively improves image quality under challenging lighting conditions. Liu et al. [19] conduct a comprehensive review and evaluation of existing LLIE algorithms and outline promising future research directions. In addition, Xiao et al. [20], [21], [22] explore deep learning-based solutions for image restoration, moiré pattern removal, and local motion deblurring, respectively, effectively addressing various challenges in image enhancement. Recently, diffusion models [23], [24], [25] have attracted increasing attention in the LLIE domain, achieving impressive results. However, most diffusion-based approaches rely on complex multi-stage training procedures, which significantly increase computational cost and implementation complexity. In contrast, our proposed DHSNet adopts a single-stage architecture, simplifying the training pipeline while delivering superior performance and strong generalization across diverse tasks.

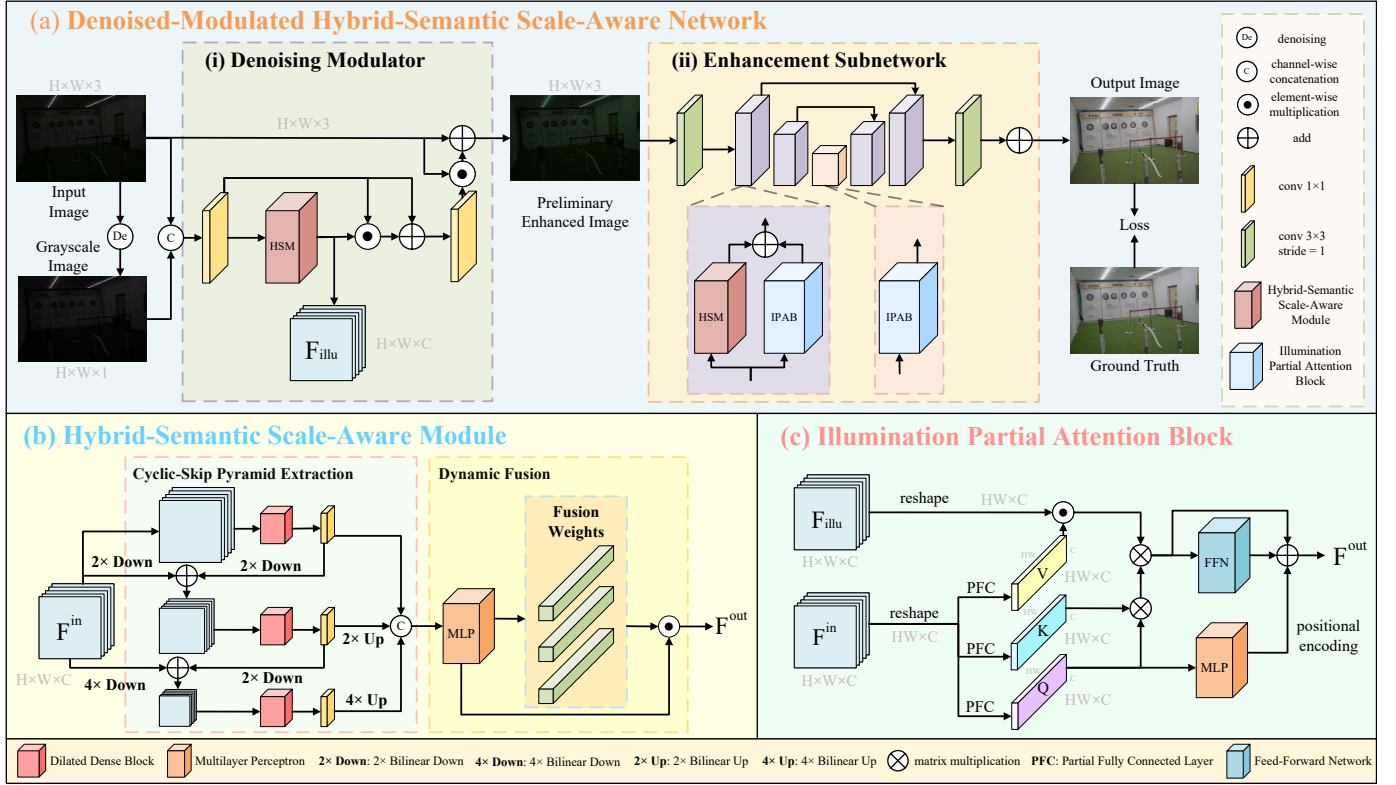


Fig. 2. Overview of DHSNet. (a) DHSNet consists of two parts: (i) the Denoising Modulator and (ii) the Enhancement Subnetwork. (b) Detailed structure of the HSM. (c) Detailed structure of the IPAB.

B. Vision Transformer

Vision Transformer (ViT) [26] pioneered the application of Transformers to computer vision by reformulating image classification as a sequence modeling task. Subsequently, numerous Transformer-based methods have been adapted for LLIE. Diverging from ViT's spatial approach, Restormer [4] employs self-attention mechanisms along the channel dimension, reducing model parameters while enhancing feature dependency modeling. SNR-Net [27] integrates SNR-aware Transformers with convolutional architectures to dynamically enhance pixels through spatially variant operations, mitigating the impact of ultra-low SNR regions. Cai et al. [5] further proposed an Illumination-Guided Transformer (IGT) that leverages lighting cues to steer non-local interactions under varying illumination conditions. However, these methods often rely on simplistic encoder-decoder structures for multi-scale feature extraction, limiting their capacity to address image corruptions across heterogeneous semantics and scales. In contrast, our approach explicitly emphasizes multi-scale feature extraction and fusion across semantic hierarchies, significantly enhancing the network's robustness to noise and artifacts during image enhancement.

III. METHOD

In this section, we will introduce the overall architecture of the DHSNet. As shown in Figure 2 (a), the Denoised-Modulated Hybrid-Semantic Scale-Aware Network (DHSNet) consists of two main modules: the Denoising Modulator and

the Enhancement Subnetwork. The Denoising Modulator is used for initial image enhancement and to improve the network's robustness to noise, while the Enhancement Subnetwork further restores noise, artifacts, and other corruptions in the image, enhances its quality, and also corrects the image illumination. The Enhancement Subnetwork consists of two key components we designed: Hybrid-Semantic Scale-Aware Module (HSM) and Illumination Partial Attention Block (IPAB). HSM extracts and fuses multi-scale features across different semantic levels, as shown in Figure 2 (b). IPAB enhances the network's nonlinear transformation ability and corrects image illumination, as shown in Figure 2 (c).

A. SNR-Based Denoising Mechanism

1) *Denoising through Image Fusion with SNR Map:* Unlike previous LLIE methods [4], [27], we improve noise robustness by performing denoising during preprocessing without introducing any trainable parameters. This not only accelerates the overall process but also provides a denoising prior that guides the enhancement network via a modulation mechanism, thereby strengthening its ability to perceive and handle noise. Following classical non-learning-based denoising approaches [28], [29], we model noise in low-light regions as pixel-level intensity jumps. Specifically, we compute a SNR map by measuring the distance between the noisy image and an estimated clean counterpart to identify noisy pixels. To suppress noise while preserving structural details, we normalize this SNR map to the range [0, 1] and derive a denoising mask. The mask

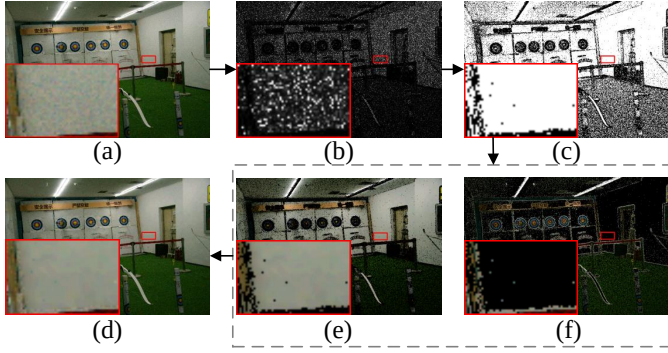


Fig. 3. The denoising process of the Denoising through Image Fusion with SNR Map is illustrated as follows: (a) shows the input image; (b) is the SNR map; (c) represents the noise mask; (d) is the denoised image. (e) and (f) display the outputs after applying the noise mask to the input image and its Gaussian-smoothed version, respectively. The lighting parameters in (a) through (f) have been manually adjusted to enhance visual contrast for better comparison of the denoising effects.

is then used to blend the original image with its Gaussian-smoothed version, effectively smoothing noisy areas while retaining fine details in non-noisy regions.

The denoising process is illustrated in Figure 3. Given an image $I \in \mathbb{R}^{H \times W \times 3}$, we first compute its Gaussian-smoothed image $G \in \mathbb{R}^{H \times W \times 3}$ and obtain the grayscale images $I_g \in \mathbb{R}^{H \times W}$ and $G_g \in \mathbb{R}^{H \times W}$ of I and G , respectively. Next, we compute the SNR map $S \in \mathbb{R}^{H \times W}$ and generate the SNR mask $S' \in \mathbb{R}^{H \times W}$. Finally, the denoised image $D \in \mathbb{R}^{H \times W}$ is obtained through a mask-guided blending operation. The computation process of Denoising through Image Fusion with SNR Map is formulated as

$$G = \text{gauss}(I) \quad (1)$$

$$I_g = 0.299 \times I_R + 0.587 \times I_G + 0.114 \times I_B \quad (2)$$

$$G_g = 0.299 \times G_R + 0.587 \times G_G + 0.114 \times G_B \quad (3)$$

$$N = |I_g - G_g|, \quad S = \frac{I_g}{(N + \epsilon)} \quad (4)$$

$$S' = 1_{S_i \geq s}, \quad i = \{1, \dots, H \times W\} \quad (5)$$

$$D = G \times S' + I \times (1 - S') \quad (6)$$

where $N \in \mathbb{R}^{H \times W}$ is the noise component, *gauss* denotes a 5×5 Gaussian filter. I_R , I_G , and I_B represent the three channels of I , while G_R , G_G , and G_B represent the three channels of G . ϵ is a small number to prevent division by zero, s is the noise threshold, and $H \times W$ is the feature map size. As shown in Figure 3 (d), the denoised image effectively removes most of the noise while preserving details.

2) *Denoising Modulator*: Although the denoising preprocessing can quickly and effectively remove local noise, its reliance on empirically set parameters may produce approximate SNR values, resulting in inaccurate noise estimation and some loss of fine details. To mitigate this issue, we modulate both the denoised output and the original input to recover details that might have been lost and to produce a preliminary enhanced image. Moreover, well-exposed regions

can supply valuable semantic context for underexposed areas, but exposure information may exist at different scales. To address this, we employ the HSM as the core component of the modulator. HSM captures and integrates multi-scale information across various semantic levels to generate a more accurate illumination map, which then guides the IPAB in correcting image illumination and improving brightness adaptation. The computation process is formulated as

$$I_{in} = \text{conv}_{1 \times 1}(\text{concat}(D_g, I)) \quad (7)$$

$$L = \text{HSM}(I_{in}) \quad (8)$$

$$I_d = \text{conv}_{1 \times 1}(L \odot I_{in} + I_{in}) \odot I + I \quad (9)$$

where $\text{conv}_{1 \times 1}$ denotes pointwise convolution, *concat* represents concatenation, D_g is the grayscale image of D , L represents the illumination feature map, and $\odot$ indicates element-wise multiplication. The output of the Modulator is denoted as I_d . This modulation mechanism enables the network to more accurately localize potential noise regions, thereby facilitating effective denoising and markedly improving the network's noise sensitivity and robustness.

It is worth noting that, although image fusion based on the SNR map can effectively suppress local noise, its denoising performance becomes limited when noise levels rise, noise occurs continuously, and noise sizes vary widely. In these cases, the SNR map is heavily disturbed and produces inaccurate estimates. To avoid losing excessive detail, the Denoising Modulator then tends to preserve the original image content rather than rely on the denoised result. To overcome this limitation, we introduce the HSM. HSM endows each layer of the subsequent enhancement subnetwork with strong restoration capabilities, allowing it to effectively address the multi-scale corruptions that remain after the preprocessing denoising stage.

B. Hybrid-Semantic Scale-Aware Module

Previous deep learning-based LLIE methods [4], [5], [6] typically use CNNs or Transformers to capture local and global information, relying on encoder-decoder architectures to provide multi-scale information for addressing noise, artifacts, and color distortion in images. However, image degradation often manifests at different semantic levels and scales across various layers of the network. If certain degradations are not effectively processed at specific layers, they may propagate to subsequent layers through skip connections. To address this issue, we design the HSM, which is aimed at better extracting and dynamically fusing multi-scale features at multiple semantic levels, thus effectively handling various types of image degradation. HSM consists of two main components: a recurrent-skip pyramid feature extractor and cross-scale dynamic fusion implemented through an inverted bottleneck structure [30], as shown in Figure 2 (b).

1) *Recurrent-Skip Pyramid Feature Extractor*: We process the input features of each layer in the pyramid using dilated dense block. Each dilated dense block consists of some dilated convolutional layers [31] with different dilation rates, which

refine the input features. Specifically, given the input features $F_i^{in} \in \mathbb{R}^{H \times W \times C}$ as F_i of the i^{th} ($i = 0, 1, 2$) layer in the pyramid, the computation of the dilated dense block is defined as

$$F_i^{out} = \text{conv}_{1 \times 1}[\text{ReLU}(\text{conv}_{3 \times 3}^d(F_i^k))], (k = 0, 1, \dots, L) \quad (10)$$

where F_i^k , ($k = 0, 1, \dots, l - 1$) denotes all intermediate features before layer l . The operator $\text{conv}_{3 \times 3}^d$ processes these concatenated features using a 3×3 convolutions with dilation rates d , and ReLU is applied as the activation function. A final 1×1 convolution is then applied to adjust the number of output channels to match F_i^{in} .

To better facilitate the flow of information, we incorporate recurrent-skip connections into the pyramid structure. Notably, the recurrent skip connections add minimal computational overhead while significantly enhancing the network's ability to capture multi-semantic and multi-scale information. The computation process can be defined as

$$F_{i+1}^{in} = \text{Down}_{2^{i+1}}(F_0^{in}) + \text{Down}_2(F_i^{out}) \quad (11)$$

where $\text{Down}_{2^{i+1}}$ denotes bilinear interpolation downsampling by a factor of 2^{i+1} .

2) *Cross-Scale Dynamic Fusion*: In order to enable interaction between multi-scale features across different semantic levels, we use a dynamic fusion module based on an inverted bottleneck structure to integrate features extracted by the pyramid module. This structure effectively prevents information loss and significantly enhances the network's expressive power. Specifically, we first apply global average pooling to the output of the pyramid extraction module to obtain a one dimensional global feature vector $v_i \in \mathbb{R}^C$. Next, a dynamic weight learning module captures the channel dependencies of the feature maps. Finally, these learned weights are used to fuse the pyramid outputs, producing the final HSM result. The process is formulated as

$$v_i = \frac{1}{HW} \sum_{s=1}^H \sum_{t=1}^W F_i^{out}(s, t) \quad (12)$$

$$[w_0, w_1, w_2] = \text{MLP}([v_0, v_1, v_2]) \quad (13)$$

$$F^{out} = \sum_{k=0}^2 w_k \cdot F_k^{out} \quad (14)$$

where s and t denote the spatial coordinates, corresponding to the row and column indices of pixels in the feature map. The *MLP* consists of three point-wise convolutions. w_0 , w_1 , and w_2 correspond to the adaptive weights for F_0^{out} , F_1^{out} , and F_2^{out} , respectively. Through cross-scale dynamic fusion, the network adjusts the fusion weights during training to capture the required features and adaptively process new images.

It is worth noting that we omit HSM from the bottom of the encoder-decoder structure to avoid excessive computational overhead. Thanks to its powerful feature extraction and fusion capabilities, HSM endows almost every network layer with strong restoration and enhancement abilities, resulting in more effective processing and higher-quality outputs. Moreover,

HSM is plug-and-play: in the denoising modulator, it also functions as a gating mechanism to generate a more accurate illumination map that guides IPAB in lighting correction. Although HSM adaptively optimizes channel weights via cross scale dynamic fusion, selectively strengthening channels that carry critical illumination information, its ability to capture long range dependencies still relies on the progressive accumulation of hierarchical implicit features within its multi layer convolutional design. In contrast, Transformer's self attention mechanism can explicitly capture long range dependencies across the entire image in a single pass, which is particularly beneficial for illumination correction and complements HSM's strengths. Therefore, we introduce the Transformer based IPAB, whose global correlation modeling capability effectively calibrates illumination distribution and further enhances correction accuracy.

C. Illumination Partial Attention Block

Traditional deep learning approaches for LLIE grounded in Retinex theory [32], [4], [5] often overlook that the scale of illumination information can vary significantly across different image regions. Moreover, relying on fully connected layers or standard convolutions to capture inter-regional correlations constrains the model's capacity for nonlinear transformation. To address these limitations, we introduce the HSM, which enables more precise extraction of illumination maps, and we propose the Partial Fully Connected Layer (PFC), designed to enhance nonlinear transformation capability while preserving original information for better generalization. In contrast to Dropout [33], our PFC retains all information rather than discarding any features. Building on these insights, we develop the IPAB, which offers stronger nonlinear transformation capability and improved generalization, allowing it to capture channel correlations more effectively for accurate illumination correction.

Given the input $F^{in} \in \mathbb{R}^{H \times W \times C}$ of IPAB, we first reshape it into a sequence of tokens $X \in \mathbb{R}^{HW \times C}$. Then, it is divided into k segments on the last dimension, representing k heads, denoted as $X = \{X_i\}_{i=1}^k$, where $X_i \in \mathbb{R}^{H \times W \times \frac{C}{k}}$. To align with the shape of X , we also reshape F_{illu} into $Y \in \mathbb{R}^{HW \times C}$ and split it into k heads. It is important to note that only the self-attention computation process of X_1 is shown in Figure 2 (c). The self-attention computation process for each head in IPAB is formulated as

$$W(X) = PFC(X) = [FC_{\frac{C}{n}}(X_{\frac{C}{n}}), X_{\frac{(n-1) \times C}{n}}] \quad (15)$$

$$Q_i = X_i W_{Q_i}^T, \quad K_i = X_i W_{K_i}^T, \quad V_i = X_i W_{V_i}^T \quad (16)$$

$$\text{Attention}_i(Q_i, K_i, V_i) = (V_i \odot Y_i) \otimes \text{softmax}\left(\frac{K_i^T Q_i}{r_i}\right) \quad (17)$$

where the *PFC* operation is defined as splitting the input X into two parts: the first $\frac{C}{n}$ portion of channels is passed through a fully connected transformation, while the remaining $\frac{(n-1) \times C}{n}$ channels are retained unchanged. X_i is the input element of the i^{th} head, Q_i , K_i , and V_i are the query, key, and value elements of the i^{th} head, respectively. $W_{Q_i}^T$, $W_{K_i}^T$, and $W_{V_i}^T$

denote the transposed matrices of the PFC layer. $\odot$ is the Hadamard product. $\otimes$ represents matrix multiplication, and r_i is a learnable parameter used for adaptive scaling of the matrix multiplication.

In summary, the proposed framework consists of three main modules: SNR-Based denoising mechanism, HSM, and IPAB. The denoising mechanism first alleviates local noise interference and, aided by HSM, generates an illumination map while effectively preprocessing and enhancing the input image. HSM not only produces a high-precision illumination map but also exploits multi-scale and multi-semantic information to perform detailed image restoration and initial illumination correction. Within IPAB, the core PFC module reduces the number of training parameters and computational cost while enhancing the model's nonlinear transformation capability. This enables IPAB to accurately weight each channel feature and guide the network's attention toward channels carrying critical information. Additionally, IPAB leverages the precise illumination map generated by HSM to perform more effective illumination correction. The collaborative operation of the SNR-Based denoising mechanism, HSM, and IPAB enables the network to achieve an optimal balance between image restoration and illumination correction, thereby further improving visual quality and overall image fidelity.

D. Loss Function

To optimize the network for LLIE tasks, we use a mixed loss function to supervise the network training. This function includes L1 loss $\mathcal{L}_1$, edge loss $\mathcal{L}_e$ [47], structural similarity loss $\mathcal{L}_s$ [48], and perceptual loss $\mathcal{L}_p$ [49]. Specifically, $\mathcal{L}_1$ ensures pixel-level fidelity, $\mathcal{L}_e$ preserves edge structures, $\mathcal{L}_s$ maintains structural consistency, and $\mathcal{L}_p$ enhances perceptual quality from a high-level feature perspective. This mixed loss function can be formulated as

$$\mathcal{L}(I_e, I_r) = \lambda_1 \cdot \mathcal{L}_1(I_e, I_r) + \lambda_p \cdot \mathcal{L}_p(I_e, I_r) + \lambda_s \cdot \mathcal{L}_s(I_e, I_r) + \lambda_e \cdot \mathcal{L}_e(I_e, I_r) \quad (18)$$

where λ_1 , λ_p , λ_s , and λ_e are the weights to balance all components of the loss function $\mathcal{L}$. I_e represents the enhanced image, and I_r represents the ground truth.

In these four loss terms, L1 loss, structural similarity loss, and edge loss typically improve PSNR and SSIM scores. These metrics are especially important for machine vision tasks, as they ensure accurate illumination and preserve fine details and textures. In contrast, the perceptual loss focuses more on human visual perception such as overall realism and visual quality and can significantly improve the LPIPS metric. Because FID primarily measures overall image distribution consistency, it is also indirectly improved by the perceptual loss. To balance pixel-level fidelity and perceptual quality, we empirically set the loss weights as follows: $\lambda_1 = 1$, $\lambda_p = 0.1$, $\lambda_s = 0.5$, and $\lambda_e = 50$.

IV. EXPERIMENTS

A. Datasets and Implementation Details

Our method was evaluated on the LOL-v1 [32], LOLv2-Real [32], LOLv2-Synthetic [35], and LSRW [50] datasets. To

further assess its performance, we conducted experiments on two challenging datasets: Sony-Total-Dark [51] and SICE [52], which includes the Grad and Mix test sets [53]. Additionally, we trained the model on LOL-v1 with varying noise levels and tested it to evaluate its noise adaptation capabilities. Finally, we assessed the generalization ability of our method on five no-reference datasets: LIME [10], NPE [54], MEF [55], DICM [56], and VV [57], as well as on ExDark [58].

1) *LOL*: The LOL dataset comprises two versions, v1 and v2, with LOL-v2 further divided into real and synthetic subsets. The training and testing sets for LOL-v1, LOv2-Real, and LOv2-Synthetic are split in the ratios of 485:15, 689:100, and 900:100, respectively. During training, we crop LOL-v1 and LOL-v2-Synthetic images into 128×128 patches and LOL-v2-Real images into 256×256 patches, using a batch size of 8.

2) *LSRW*: The LSRW dataset contains 5,650 image pairs captured from various scenes. Among these, 5,600 pairs are randomly selected for training, and the remaining 50 pairs are used for evaluation.

3) *Sony-Total-Dark*: This dataset is a customized subset of See-in-the-Dark (SID) [51] and comprises 2697 short and long exposure RAW image pairs. Of these, 2099 pairs are used for training and 598 for testing. To more effectively assess the model's performance under extremely low-light conditions, the RAW format images are converted to sRGB without gamma correction, resulting in significantly darker images. During training, each image is cropped into 256×256 patches, with a batch size of 4.

4) *SICE*: The SICE dataset contains 589 images featuring both low-light and overexposed scenes, which we divide into training, validation, and testing subsets in a 7:1:2 proportion. For training, we randomly crop 160×160 patches from the training images and set the batch size to 10. To further evaluate the model's performance under dynamic lighting conditions, we test it on the unseen SICE-Grad and SICE-Mix datasets.

5) *LOL-v1 with Varying Noise Levels*: To evaluate the model's performance across a range of noise intensities, from low to high, we artificially added various levels of Gaussian noise to the LOL-v1 dataset. Specifically, at each pixel location, the noise follows a Gaussian distribution with mean -0.05 and a standard deviation controlled jointly by the parameter ϵ and the maximum standard deviation σ_{max} . This simulates the slightly dark noise typically seen under low light conditions. To ensure a natural appearance, the added noise is applied to the entire image rather than only to its dark regions. Mathematically, this process can be expressed as

$$I_n(s, t) = I(s, t) + n(s, t) \quad (19)$$

$$n(s, t) \sim \mathcal{N}(\mu = -0.05, \sigma = \min(\epsilon, \sigma_{max})) \quad (20)$$

where $I(s, t)$ and $I_n(s, t)$ denote the pixel intensities of the original and noisy images at location (s, t) , respectively, and $n(s, t)$ is the corresponding Gaussian noise. The larger the ϵ value, the higher the noise level. Finally, we generated four versions of the LOL-v1 dataset with noise intensities of $\epsilon = 0.1, 0.25, 0.5$, and 1.0 , respectively.

TABLE I

QUANTITATIVE COMPARISONS ON THE LOL-v1 AND LOL-v2 DATASETS, WHERE INFERENCE SPEED AND FLOPs ARE MEASURED USING A SINGLE 256×256 IMAGE, AND THE BEST AND SECOND-BEST RESULTS ARE MARKED IN RED AND BLUE, RESPECTIVELY.

Method	Complexity			LOL-v1				LOv2-Real				LOv2-Synthetic			
	Speed (s)↓	Params (M)↓	FLOPs (G)↓	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓
UFormer [34]	0.268	5.29	12.00	16.36	0.771	0.354	109.35	18.82	0.771	0.347	98.13	19.66	0.871	0.227	74.76
RetinexNet [32]	0.841	0.84	587.47	16.77	0.560	0.311	126.26	15.47	0.567	0.372	133.90	17.13	0.798	0.259	89.14
Sparse [35]	0.059	2.33	53.26	17.20	0.640	0.232	97.15	20.06	0.816	0.214	84.21	22.05	0.905	0.160	50.37
EnGAN [36]	2.254	114.35	61.01	17.48	0.650	0.203	94.70	18.23	0.617	0.220	84.04	16.57	0.734	0.264	103.81
RUAS [37]	0.015	0.003	0.83	18.23	0.720	0.191	101.97	18.37	0.723	0.198	94.16	16.55	0.652	0.237	97.66
FIDE [38]	0.043	8.62	18.27	18.27	0.665	0.173	95.86	16.85	0.678	0.251	120.07	15.20	0.612	0.238	107.58
PairLIE [39]	0.081	0.33	20.81	19.51	0.736	0.248	98.13	19.89	0.778	0.317	61.29	19.07	0.794	0.230	67.56
DRBN [40]	2.462	5.27	20.13	20.13	0.830	0.159	98.73	20.29	0.831	0.268	89.08	23.22	0.927	0.114	28.74
KinD [41]	2.130	8.02	34.99	20.86	0.790	0.146	78.28	18.06	0.825	0.151	95.02	17.41	0.806	0.255	97.32
GlobalDiff [23]	1.218	17.36	442.02	22.02	0.848	0.137	75.63	20.15	0.846	0.113	51.72	24.47	0.929	0.051	20.17
Restormer [4]	0.114	26.13	144.25	22.43	0.823	0.147	72.99	19.94	0.827	0.232	58.64	21.41	0.830	0.144	32.81
LLFormer [42]	0.112	22.52	24.55	23.65	0.816	0.153	75.42	20.10	0.792	0.182	84.73	24.04	0.909	0.073	27.44
MIRNet [43]	0.205	31.76	785.00	24.14	0.830	0.139	68.95	20.02	0.820	0.199	82.25	21.94	0.876	0.120	40.18
SNR-Net [27]	0.027	4.01	26.35	24.61	0.842	0.151	55.12	21.48	0.849	0.157	54.87	24.14	0.928	0.056	19.88
Retinexformer [5]	0.012	1.61	15.57	25.16	0.845	0.131	69.36	22.80	0.840	0.171	59.27	25.67	0.930	0.059	22.51
Ours	0.018	3.16	36.94	25.04	0.868	0.081	36.59	23.97	0.886	0.091	41.44	27.94	0.959	0.027	10.94

TABLE II

QUANTITATIVE COMPARISONS USING THE GT MEAN STRATEGY ON THE LOL-v1 AND LOL-v2 DATASETS, WITH THE BEST AND SECOND-BEST RESULTS MARKED IN RED AND BLUE, RESPECTIVELY. '-' INDICATES THAT THE MODEL DID NOT PROVIDE THE RELEVANT CODE OR EXPERIMENTAL RESULTS. RETINEXFORMER_ IS A VARIANT OF RETINEXFORMER THAT INCORPORATES THE GT MEAN STRATEGY.

Method	Complexity			LOL-v1				LOv2-Real				LOv2-Synthetic			
	Speed (s) ↓	Params (M) ↓	FLOPs (G) ↓	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓	PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓
LLFlow [44]	0.077	38.86	26.31	25.13	0.872	0.117	65.17	26.20	0.888	0.137	70.68	24.81	0.919	0.067	20.24
Retinexformer_ [5]	0.012	1.61	15.57	27.18	0.850	0.127	69.82	27.71	0.856	0.166	59.17	29.04	0.939	0.055	23.15
GlobalDiff [23]	1.218	17.36	442.02	27.84	0.877	0.091	47.43	28.82	0.895	0.095	53.18	28.67	0.944	0.047	17.29
DiffLL [45]	0.226	22.08	-	26.33	0.845	0.217	48.11	28.66	0.876	0.207	45.36	28.87	0.884	0.047	18.35
ExpoMamba [46]	0.010	41.00	-	25.77	0.860	0.212	-	28.04	0.868	0.203	-	-	-	-	-
GLARE [25]	-	59.45	508.42	27.35	0.883	0.083	-	28.98	0.905	0.097	-	29.84	0.958	-	-
Ours	0.018	3.16	35.97	27.27	0.891	0.076	36.62	29.31	0.909	0.083	41.45	32.14	0.969	0.024	10.96

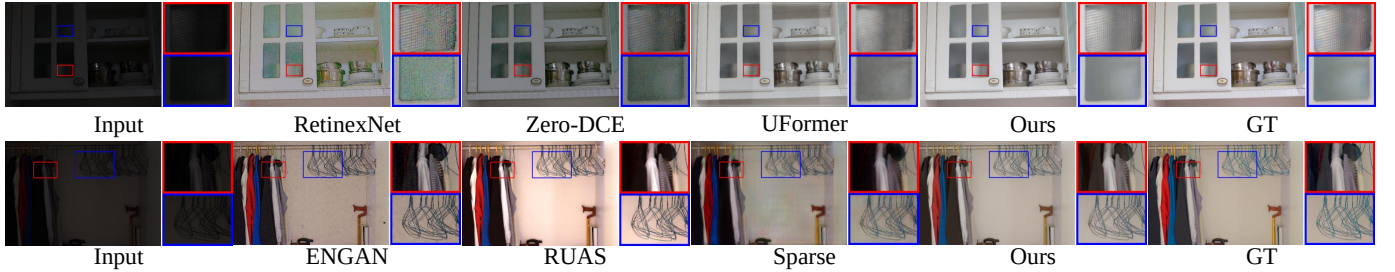


Fig. 4. Visual results of different methods on the LOL-v1 dataset.

6) *No-Reference Datasets*: To evaluate the model's generalization ability, following the DiffLL [45] setting, all models trained on the LOL-v1 dataset were tested on DICM, LIME, MEF, NPE, and VV.

7) *ExDark*: The ExDark dataset contains 7,363 low-light images across 12 object categories, ranging from twilight to extreme darkness. We split the dataset into 80% for training and 20% for testing. YOLOv3 [70] was trained on the enhanced ExDark dataset using various LLIE models. All images were cropped to 608×608 pixels, and training was conducted for 25 epochs using the SGD optimizer with a batch size of 8.

8) *Experiment Settings*: DHSNet is implemented in PyTorch and trained using the Adam [71] optimizer ($\beta = 0.9$ and $\beta = 0.999$) on a single GeForce RTX 3090 GPU. The initial learning rate is set to 2×10^{-4} , with a weight decay of 1×10^{-4} , and is steadily decreased via a cosine annealing schedule [72] during training. Data augmentation includes random horizontal and vertical flips.

9) *Evaluation Metrics*: For the LOL, LSRW, and LOL-v1 datasets with varying noise levels, we use Peak Signal-to-Noise Ratio (PSNR), Structural Similarity Index (SSIM) [48], Learned Perceptual Image Patch Similarity (LPIPS) [73], and Fréchet Inception Distance (FID) [74]

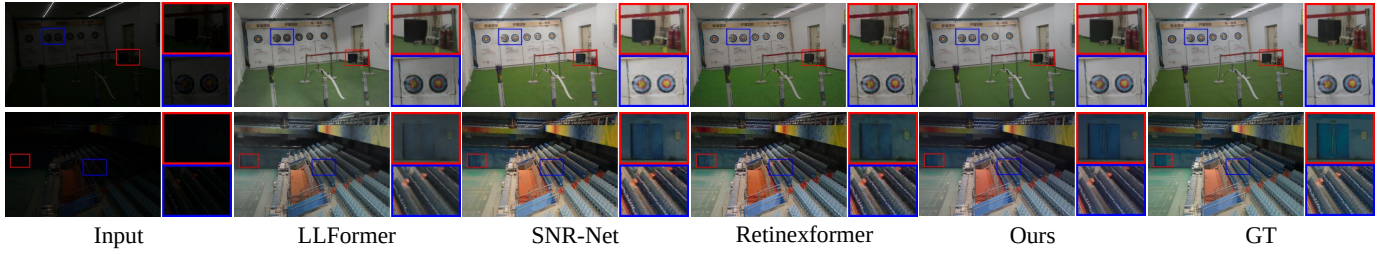


Fig. 5. Visual comparisons of different methods on the LOLv2-Real dataset

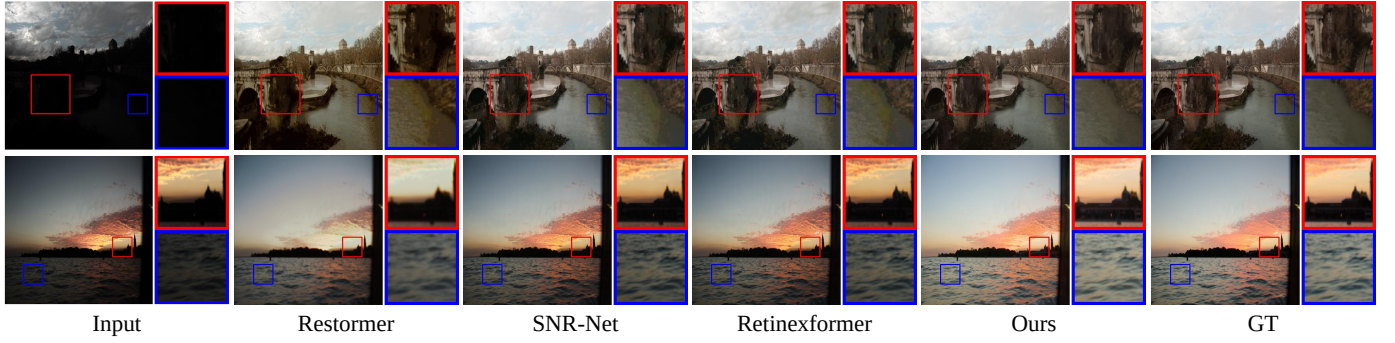


Fig. 6. Visual results of different methods on LOLv2-Synthetic dataset.



Fig. 7. Visual comparisons of different methods on the Sony-Total-Dark dataset under extremely low-light conditions.

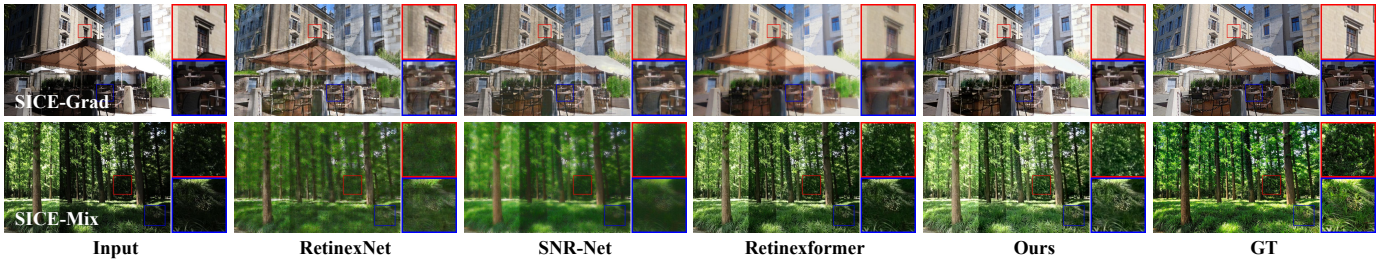


Fig. 8. Visual results of different methods on the SICE-Grad and SICE-Mix subsets under dynamic lighting conditions.

as evaluation metrics. For the Sony-Total-Dark and SICE datasets, we use PSNR and SSIM. These metrics comprehensively assess restoration quality and its alignment with human visual perception. For the five no-reference datasets, we use NIQE [75], BRISQUE [76], and PI [77] to comprehensively assess image quality, model stability, and generalization capability in the absence of ground-truth references. For the ExDark dataset, we use AP, mAP@0.5, and mAP@[0.5:0.95] to comprehensively evaluate the performance of the enhanced images under unseen scenarios in downstream object detection tasks.

B. Quantitative and Qualitative Results

1) *Results on LOL and LSRW Datasets:* When evaluating on the LOL dataset, existing LLIE models can be divided into two categories based on whether they use the GT Mean strategy (i.e., using Ground Truth to correct the output). To ensure a fair comparison, for models that do not use the GT Mean strategy, our DHSNet also does not use it, with results shown in Table I. Conversely, when compared to models that do employ the GT Mean strategy, our DHSNet likewise adopts the GT Mean strategy, with results shown in Table II. Experimental results across multiple metrics demonstrate that our

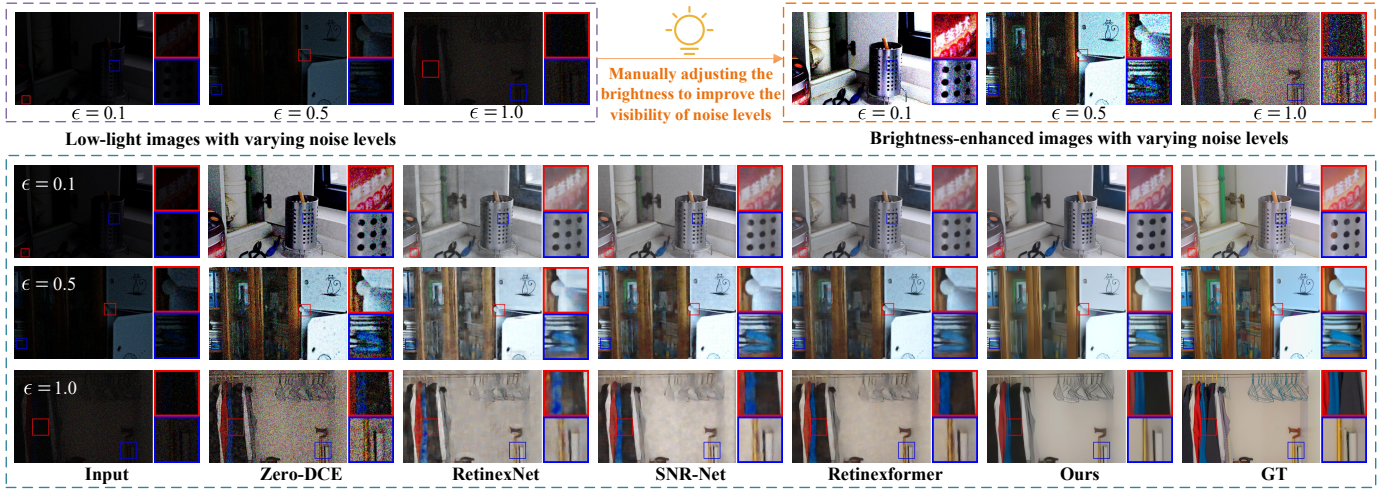


Fig. 9. Visual comparisons of different methods on the LOL-v1 dataset with varying levels of added noise. The first row shows images with different intensities of Gaussian noise, along with their corresponding versions after manual illumination adjustment to better visualize noise levels. The second to fourth rows present enhancement results of different methods under progressively increasing noise conditions.

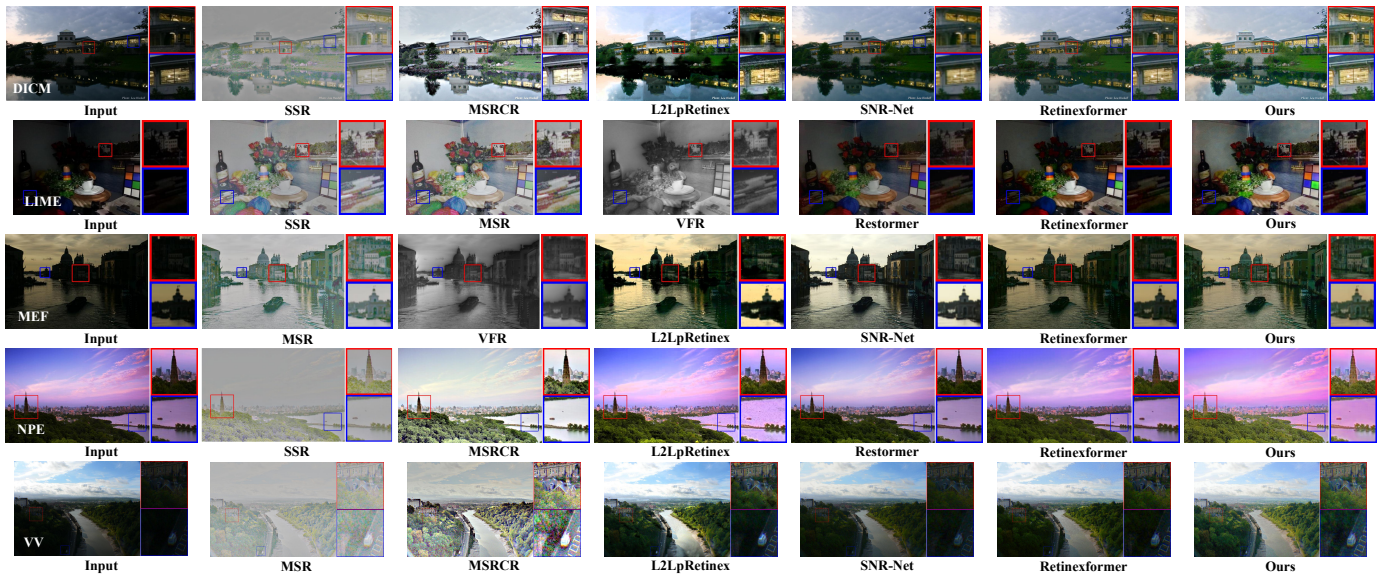


Fig. 10. Visual results on the five no-reference datasets: DICM, LIME, MEF, NPE, and VV.

method produces higher-quality images. Moreover, DHSNet strikes a favorable balance in computational complexity and achieves very fast inference speed. Visual comparisons between different methods are shown in Figure 4, Figure 5, and Figure 6. Benefiting from the denoising mechanism, HSM, and IPAB, our method outperforms other SOTA approaches in visual quality, with more accurate details, consistent colors, and enhanced brightness that better conform to human visual perception. The quantitative comparison results on the LSRW dataset are presented in Table III. The experimental results demonstrate that DHSNet outperforms other methods across multiple metrics, highlighting its strong capability to handle low-light conditions in real scenes and deliver superior image enhancement.

2) *Results on Sony-Total-Dark and SICE Datasets:* To validate the performance of DHSNet on large-scale datasets and under complex lighting conditions, we evaluate it on

the Sony-Total-Dark and SICE datasets, testing on the unseen Grad and Mix subsets. As shown in Table IV, our method achieves the best performance across all metrics on both datasets, demonstrating the effectiveness of DHSNet in extremely low-light and dynamically varying illumination environments. Qualitative comparisons are presented in Figure 7 and Figure 8. Benefiting from the powerful restoration capabilities of HSM during the enhancement process, our method generates fewer artifacts and preserves finer details. Moreover, in some severely blurred low-light scenarios, the enhanced images appear even clearer than the ground truth. Moreover, with the precise illumination correction enabled by IPAB, the enhanced images exhibit lighting that more closely aligns with human visual perception.

3) *Results on LOL-v1 Dataset with Varying Noise Levels:* Quantitative results of DHSNet under different Gaussian noise levels are shown in Table V. The experiments demonstrate

TABLE III
QUANTITATIVE COMPARISONS ON THE LSRW DATASET, WITH THE BEST AND SECOND-BEST RESULTS MARKED IN RED AND BLUE, RESPECTIVELY.

Method	Speed (s) ↓	LSRW			
		PSNR ↑	SSIM ↑	LPIPS ↓	FID ↓
SNR-Net [27]	0.027	16.50	0.505	0.419	65.81
Uformer [34]	0.268	16.59	0.494	0.435	82.30
Restormer [4]	0.114	16.30	0.453	0.427	59.22
Palette [59]	0.056	13.57	0.476	0.479	73.84
UHDFour _{2x} [60]	0.069	17.30	0.529	0.443	52.03
WeatherDiff [61]	1.757	16.50	0.487	0.431	96.05
GDP [62]	-	12.88	0.362	0.412	76.91
DiffLL [45]	0.052	19.28	0.552	0.350	45.29
Ours	0.018	19.89	0.593	0.331	42.64

TABLE IV
QUANTITATIVE RESULTS OF NO-REFERENCE EVALUATION ON THE SONY-TOTAL-DARK AND SICE DATASETS, WITH THE BEST-PERFORMING VALUES HIGHLIGHTED IN RED.

Methods	Speed (s) ↓	Sony-Total-Dark		SICE	
		PSNR ↑	SSIM ↑	PSNR ↑	SSIM ↑
RetinexNet [32]	0.841	15.69	0.395	12.42	0.613
ZeroDCE [63]	0.003	14.08	0.090	12.45	0.639
URetinexNet [64]	0.024	15.51	0.323	10.89	0.605
RUAS [37]	0.015	12.62	0.081	8.65	0.494
LLFlow [44]	0.077	16.22	0.367	12.73	0.617
SNR-Net [27]	0.027	22.41	0.660	12.66	0.632
Retinexformer [5]	0.012	22.72	0.671	12.68	0.652
Ours	0.018	23.61	0.694	12.82	0.660

that DHSNet maintains high performance across various noise levels, consistently achieving SOTA metrics. This success stems from two key components: the denoising mechanism, which effectively preprocesses localized inherent noise, and HSM, which provides robust restoration of large, continuous noise patterns through multi-scale, multi-semantic feature capture. Qualitative comparisons are presented in Figure 9. Even when noise severely degrades image regions and details are largely lost, our method preserves most critical information and removes nearly all noise, resulting in enhanced images that better align with human visual perception and deliver superior detail restoration and illumination correction.

4) *Results on No-Reference Datasets*: To validate the generalization ability of our method across diverse real-world scenarios and different image sizes, we compared our method against Retinex-based traditional approaches and various deep learning models on five no-reference datasets, as shown in Table VI. DHSNet achieves outstanding NIQE results, showing effective noise removal and contrast enhancement. It also scores low on BRISQUE, indicating better preservation of details and textures, and excels on the PI metric, reflecting outputs that closely match human visual perception. Qualitative comparison results are shown in Figure 10. It is evident that our method produces clearer detail restoration, more thorough noise suppression, and more accurate illumination correction,

resulting in enhanced images that are visually superior to those of other methods.

5) *Results on ExDark Dataset*: We performed detection on images processed by various LLIE methods and compared the results to evaluate the performance of enhanced images in high-level vision tasks. The quantitative experimental results in Table VII show that our method outperforms the baseline models and other LLIE approaches in overall detection performance across all 12 categories. Figure 11 illustrates YOLO-v3's detection outputs on ExDark dataset images enhanced by different methods. DHSNet excels in brightness adjustment, color restoration, and detail preservation. The images it generates closely resemble real-world scenes, leading to more accurate object localization and identification.

C. Ablation Study

To validate the effectiveness of our approach, we conducted ablation experiments on multiple datasets. The results are summarized in Tables VIII, IX, and X.

1) *Network Structure*: We conducted ablation experiments on the key modules of DHSNet using the LOL-v1 dataset, see Table VIII. The results show that removing any module leads to a drop in performance. Removing the denoising modulator causes PSNR to fall by 0.76 dB because the network can no longer effectively repair local damage and surrounding pixels lose detail. Removing IPAB causes PSNR to decrease by 0.46 dB since a single-branch HSM is unable to capture the most direct global illumination cues needed for accurate lighting correction. Removing HSM results in a 0.84 dB drop in PSNR because using IPAB alone prevents precise estimation of the illumination map and severely limits multi-scale damage repair. Serially combining IPAB and HSM, whether HSM → IPAB or IPAB → HSM, also reduces performance. This further validates the effectiveness of our parallel dual-branch design. HSM and IPAB complement each other and increasing network width rather than depth not only improves enhancement quality but also speeds up inference to better meet real-time requirements.

2) *Hybrid-Semantic Scale-Aware Module*: We performed ablation experiments on each component of HSM using the LOLv2-Synthetic dataset to assess their individual contributions. As shown in Table IX, replacing the pyramid feature extraction module with a simple convolutional block causes PSNR to drop by 1.32 dB, and removing the recurrent-skip connections leads to a 0.64 dB decline. These results demonstrate that rich multi-scale and multi-semantic information is essential for the network's performance. Eliminating the feature fusion stage implemented via the inverted bottleneck architecture results in a 0.37 dB decrease, underscoring its role in effectively combining features and capturing their interdependencies.

3) *Model Hyperparameters*: To investigate the impact of hyperparameters on model performance, we conducted ablation experiments on the two core modules HSM and IPAB using the LOLv2-Synthetic dataset, with results shown in Table X. The experiments indicate that setting the number of output channels per layer in HSM's dense convolution blocks

TABLE V

THE RESULTS OF THE ABLATION EXPERIMENTS CONDUCTED ON THE LOL-V1 AND LOLV2-REAL DATASETS FOR THE DENOISING MODULATOR.

Methods	$\epsilon = 0.1$				$\epsilon = 0.25$				$\epsilon = 0.5$				$\epsilon = 1$			
	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	FID $\downarrow$
RetinexNet [32]	17.95	0.665	0.380	237.05	18.24	0.657	0.397	261.34	18.21	0.631	0.457	314.71	17.82	0.587	0.549	389.37
ZeroDCE [63]	12.40	0.256	0.662	204.84	12.03	0.188	0.855	225.96	11.18	0.099	0.653	306.09	10.18	0.057	1.250	386.86
SNR-Net [27]	20.30	0.719	0.333	189.28	20.59	0.699	0.358	208.99	20.46	0.660	0.421	280.60	19.49	0.622	0.501	371.62
Retinexformer [5]	21.40	0.736	0.317	186.33	21.97	0.724	0.332	194.09	21.36	0.703	0.367	249.22	20.43	0.656	0.471	325.48
Ours	22.06	0.781	0.218	99.58	22.56	0.775	0.239	119.67	21.95	0.734	0.320	170.92	21.14	0.702	0.407	239.32

TABLE VI

QUANTITATIVE RESULTS OF NO-REFERENCE EVALUATION ON DICM, LIME, MEF, NPE, AND VV DATASETS, WITH THE BEST AND SECOND-BEST RESULTS MARKED IN RED AND BLUE, RESPECTIVELY.

Method	DICM			LIME			MEF			NPE			VV			Average		
	NIQE $\downarrow$	BRISQUE $\downarrow$	PI $\downarrow$	NIQE $\downarrow$	BRISQUE $\downarrow$	PI $\downarrow$	NIQE $\downarrow$	BRISQUE $\downarrow$	PI $\downarrow$	NIQE $\downarrow$	BRISQUE $\downarrow$	PI $\downarrow$	NIQE $\downarrow$	BRISQUE $\downarrow$	PI $\downarrow$	NIQE $\downarrow$	BRISQUE $\downarrow$	PI $\downarrow$
SSR [65]	4.026	25.973	3.360	3.910	17.206	3.100	3.476	13.933	2.809	4.368	20.339	3.144	2.921	20.086	3.162	3.740	19.507	3.115
MSRCR [66]	3.642	24.168	3.385	3.753	18.264	2.840	3.280	15.217	2.388	4.401	17.206	3.183	2.543	24.455	2.880	3.524	19.862	2.935
MSR [67]	3.897	26.161	3.266	3.882	17.806	2.976	3.450	14.428	2.643	4.399	19.094	3.017	2.883	20.464	3.161	3.702	19.591	3.013
VFR [68]	4.371	28.688	4.101	4.228	17.388	3.710	3.950	21.719	3.989	5.053	27.311	4.208	3.579	25.070	4.036	4.236	24.035	4.009
L2LpRetinex [69]	3.701	17.921	3.075	3.559	16.203	3.057	3.678	15.126	2.708	3.986	14.253	2.863	2.866	17.704	3.146	3.558	16.041	2.970
DRBN [40]	4.369	30.708	3.800	4.869	44.669	4.711	4.562	34.564	3.973	3.921	25.336	3.267	3.671	24.945	3.117	4.278	32.045	3.774
Zero-DCE [63]	3.951	23.350	3.149	3.500	29.359	2.989	4.379	26.054	3.239	3.826	21.835	2.918	5.080	21.835	3.307	4.147	24.487	3.120
MIRNet [43]	4.021	22.104	3.691	4.202	34.499	3.756	4.378	28.623	3.398	3.810	23.658	3.205	3.548	22.897	2.911	3.992	26.356	3.392
EnGAN [43]	3.832	19.129	3.256	3.556	26.799	3.270	4.249	22.664	3.381	3.775	21.157	2.953	3.689	14.153	2.749	3.820	20.780	3.122
RUAS [37]	7.306	46.882	5.700	5.435	42.120	4.921	5.322	34.880	4.581	7.198	48.976	5.651	4.987	35.882	4.329	6.050	41.748	5.036
UFormer [34]	3.847	19.657	3.180	3.935	25.240	3.582	4.300	21.874	3.565	3.510	16.239	2.871	3.274	19.491	2.618	3.773	20.500	3.163
Restormer [4]	3.964	19.474	3.152	3.815	25.322	3.436	4.365	22.931	3.292	3.729	16.668	2.862	3.795	16.974	2.712	3.934	20.274	3.091
SNR-Net [27]	3.804	19.459	3.285	4.063	28.331	3.753	4.597	29.023	3.677	3.940	28.419	3.278	3.761	23.672	2.903	4.033	25.781	3.379
DiffLL [45]	3.806	18.584	3.011	3.427	24.165	3.011	3.777	19.843	3.074	3.427	15.789	2.597	3.507	14.644	2.562	3.589	18.605	2.851
Retinexformer [5]	3.625	15.467	3.141	4.080	14.189	3.103	3.433	14.797	2.716	4.090	12.063	3.002	3.955	15.232	3.345	3.837	14.350	3.061
Ours	3.422	13.754	2.916	3.707	12.308	2.895	3.259	11.784	2.600	3.706	11.861	2.701	2.766	13.589	3.060	3.372	12.659	2.834

TABLE VII

QUANTITATIVE COMPARISONS ON THE EXDARK DATASET, WITH THE BEST AND SECOND-BEST RESULTS MARKED IN RED AND BLUE, RESPECTIVELY.

Model	Bicycle	Boat	Bottle	Bus	Car	Cat	Chair	Cup	Dog	Motorbike	People	Table	mAP@0.5 $\uparrow$	mAP@[0.5, 0.95] $\uparrow$
YOLOv3 [70]	81.8	78.6	77.3	91.7	84.3	74.7	72.8	77.2	79.2	75.1	80.6	55.8	77.4	43.6
KIND [41]	80.6	78.2	77.7	94.3	84.4	67.4	69.2	77.9	79.8	75.8	81.4	54.3	76.7	42.7
Zero-DCE [63]	84.1	77.6	78.3	93.1	83.7	70.3	69.8	77.6	77.4	76.3	81.0	53.6	76.9	43.0
Restormer [4]	81.3	76.5	78.0	92.3	84.1	71.2	71.3	79.1	79.7	75.1	81.8	55.5	77.2	43.3
LLFormer [42]	80.4	76.8	77.9	92.4	83.6	73.1	71.8	78.7	79.8	76.0	82.3	56.3	77.4	43.5
SNR-Net [27]	79.1	81.1	77.7	92.6	82.8	71.1	72.7	80.1	78.7	76.0	81.7	57.4	77.6	44.0
Retinexformer [5]	81.9	77.1	77.0	93.1	84.3	72.1	71.4	78.1	79.8	78.4	82.6	57.8	77.8	43.8
Ours	83.6	78.6	77.6	93.2	84.7	71.7	73.6	79.8	79.1	76.6	83.0	58.1	78.3	44.4

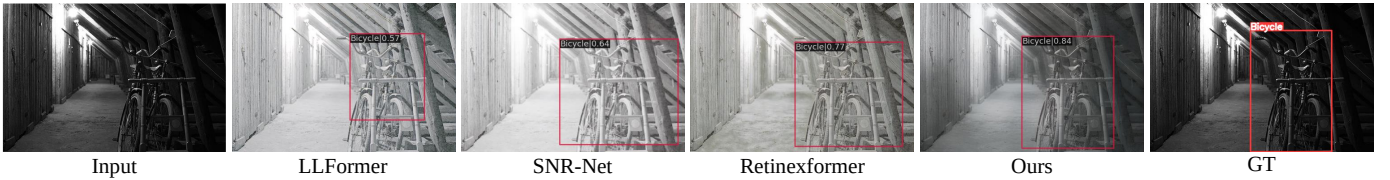


Fig. 11. Visual comparison of different methods for low-light image enhancement and object detection results on ExDark dataset.

to 20 significantly degrades performance because the network struggles to process complex multi-scale and multi-semantic features. Increasing this channel count to 80 improves performance but also raises computational complexity. In the case of IPAB, configuring the channel attention mechanism to use only 20 channels prevents the model from effectively capturing inter-channel dependencies and extracting critical information.

Raising the channel count to 80 enriches feature representation yet incurs a sharp increase in computational cost and a slight performance drop, likely due to higher overfitting risk and greater optimization difficulty. Furthermore, Table X shows that increasing HSM's channel count adds far fewer parameters and computations compared to the same increase in IPAB, since HSM relies on efficient convolution operations while

TABLE VIII
ABLATION STUDY RESULTS ON THE LOL-v1 DATASET FOR DHSNET.

Methods	Complexity			LOL-v1	
	Speed (s) ↓	Params (M) ↓	FLOPs (G) ↓	PSNR ↑	SSIM ↑
DHSNet w/o Denoising	0.016	2.89	27.92	24.14	0.847
DHSNet w/o HSM	0.010	0.97	9.31	24.20	0.851
DHSNet w/o IPAB	0.009	2.79	30.60	24.58	0.859
DHSNet w/o PFC	0.018	3.46	39.81	24.75	0.864
HSM → IPAB	0.020	3.16	36.94	24.45	0.854
IPAB → HSM	0.020	3.16	36.94	24.66	0.862
DHSNet	0.018	3.16	36.94	25.04	0.868

TABLE IX
RESULTS OF ABLATION EXPERIMENTS ON THE LOLV2-SYNTHETIC DATASET FOR HSM

Methods	Complexity			LOLv2-Synthetic	
	Speed (s) ↓	Params (M) ↓	FLOPs (G) ↓	PSNR ↑	SSIM ↑
DHSNet w/o Pyramid extractor	0.014	1.54	30.14	26.62	0.933
DHSNet w/o Recurrent-skip connections	0.018	3.16	36.96	27.30	0.948
DHSNet w/o Inverted bottleneck	0.018	3.16	36.94	27.57	0.951
DHSNet	0.018	3.16	36.94	27.94	0.959

TABLE X
THE RESULTS OF THE ABLATION EXPERIMENTS ON NETWORK ARCHITECTURE HYPERPARAMETERS CONDUCTED ON THE LOLV2-SYNTHETIC DATASET.

Methods	value	Complexity			LOLv2-Synthetic	
		Speed (s) ↓	Params (M) ↓	FLOPs (G) ↓	PSNR ↑	SSIM ↑
HSM Channel Count	20	0.016	2.17	20.16	27.31	0.953
	40	0.018	3.16	36.94	27.94	0.959
	80	0.024	6.11	88.65	28.04	0.960
IPAB Channel Count	20	0.016	0.79	9.41	26.85	0.952
	40	0.018	3.16	36.94	27.94	0.959
	80	0.038	12.57	146.36	27.09	0.950

channel attention must perform pairwise calculations across all channels. Balancing efficiency and performance, we set both HSM and IPAB to 40 channels, achieving the optimal trade-off.

V. CONCLUSION

In this work, we tackled the challenges of Low-Light Image Enhancement (LLIE) by proposing DHSNet, a novel one-stage framework that combines robust noise suppression, multi-scale feature extraction and efficient illumination correction to improve image quality. By integrating a SNR-based denoising mechanism, a Hybrid-Semantic Scale-Aware Module (HSM) and an Illumination Partial Attention Block (IPAB), DHSNet effectively reduces noise, preserves fine details and restores degraded structures. Extensive experiments on multiple LLIE datasets show that DHSNet consistently outperforms SOTA methods in both qualitative and quantitative evaluations. Its robust design and strong generalization also lead to superior results in no-reference enhancement and low-light object detection tasks, offering clear benefits for downstream vision applications.

Although DHSNet delivers excellent results in a wide range of scenarios, extremely complex lighting conditions still present challenges. In the future, we will continue to refine DHSNet and adapt it for real-time processing in environments

with complex illumination. Our goal is to boost its flexibility and robustness in dynamic settings, ensuring it meets the demands of high-performance, real-world applications.

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